WORKING PAPER — PRELIMINARY STUDY

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Continuous Health Scoring for Hyperparameter Optimization:

Addressing Hyperband's False Positive and Slow Starter Failure Modes

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Abstract

Hyperband (Li et al., 2018) improves hyperparameter optimization efficiency by eliminating underperforming configurations at fixed checkpoints based on comparative ranking. It has two well-documented failure modes: false positives (configurations that rank well at a checkpoint but deteriorate afterward) and slow starters (configurations that rank poorly early but improve later). Both occur because Hyperband makes binary decisions at fixed intervals based on current performance level, not on improvement trajectory or health trend.

We propose complementing Hyperband with continuous health scoring via the stochastic power metric $P(t) = E(t) \times W(t)$, which measures improvement velocity relative to an adaptive expected trajectory. Applied to real Pythia scaling suite data (Biderman et al. 2023), treating 7 Pythia models (70M–12B) as HPO configurations (eta=3, rungs at checkpoints 3 and 6), the combined HB+PM approach at $\theta=0.5$ recovers the 1.4B slow starter that Hyperband eliminates at Rung 1 while correctly eliminating genuinely weak configurations. The 1.4B model improves +7.1% after Rung 1, reaching a final score 0.111 above the elimination cutoff — the largest gap in the search. The power metric catches them because it tracks improvement rate, not current level. These findings are preliminary and invite collaboration on validation using standard HPO benchmarks.

Keywords: hyperparameter optimization, Hyperband, successive halving, early stopping, slow starters, health scoring, power metric, adaptive benchmarks, HPO quality

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1. Introduction

Hyperparameter optimization (HPO) is one of the most compute-intensive tasks in modern machine learning. Identifying the optimal learning rate, architecture depth, regularization strength, and other hyperparameters for a given model requires training many configurations and

comparing their performance. Naïve grid search or random search allocates compute uniformly across all configurations regardless of their performance trajectory.

Hyperband (Li et al., 2018) addresses this by early stopping: configurations are evaluated at successive checkpoints (rungs), and the worst-performing fraction is eliminated at each rung based on comparative ranking. The surviving configurations receive more compute. This achieves an order-of-magnitude speedup over random search on standard benchmarks.

Li et al. themselves acknowledge two limitations in the original paper. First, Hyperband can prematurely eliminate slow starters: configurations that perform poorly early but improve significantly with more training. Second, configurations that look strong at a rung checkpoint may subsequently deteriorate. Neither failure mode can be addressed by the rung-based ranking mechanism alone, because both require information about the trend between checkpoints rather than the value at a checkpoint.

This paper proposes a complementary mechanism: continuous health scoring via the stochastic power metric, applied between Hyperband rung checkpoints. The power metric $P(t) = E(t) \times W(t)$ tracks whether each configuration is improving faster or slower than its own adaptive expected trajectory — a signal orthogonal to Hyperband's comparative ranking. Adding this signal does not reduce Hyperband's compute savings; it improves the quality of configurations that survive the search.

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2. Hyperband and Its Failure Modes

2.1 Hyperband Mechanics

Hyperband runs multiple instances of Successive Halving with different budgets-per-configuration tradeoffs. Each instance proceeds through rounds: all N configurations run for r_1 steps, the bottom $(1-1/\eta)$ fraction are eliminated, and the survivors receive $\eta\times$ more compute in the next round. With default $\eta=3$, each round eliminates two-thirds of remaining configurations.

The elimination criterion is purely comparative: which configurations have the lowest validation loss at the current checkpoint? This ranking is taken as a proxy for final performance. The key assumption is that early performance is positively correlated with final performance. When this assumption holds, Hyperband is highly efficient. When it fails, Hyperband makes costly errors.

2.2 Two Failure Modes

Table 1. Hyperband Failure Modes and Power Metric Response

Failure Mode	What Happens in Hyperband	Power Metric Response
False Positive (Type B)	Config looks strong at Rung 1. HB keeps it. Config then deteriorates. HB wastes steps 6-20 on it.	P(t) detects declining health between rungs. Flags config early. Saves wasted compute.
Slow Starter (Type C)	Config starts weak. HB eliminates at Rung 1. Config would have improved significantly.	E(t) tracks improvement velocity, not level. Sees rising trajectory. Keeps config full run.

Note: Type B = false positive (strong early, deteriorates). Type C = slow starter (weak early, improves). Both arise when the assumption of early-final correlation fails.

The slow starter problem is particularly costly because it is an irreversible error: once a configuration is eliminated, it cannot be recovered within the current search. If the best hyperparameter combination requires more steps to manifest its quality — for example, a configuration with a longer learning rate warmup or a regularization schedule that activates later — Hyperband will never find it.

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3. The Power Metric as Complementary Signal

3.1 What the Power Metric Measures

The power metric P(t) = E(t) × W(t) measures two properties of a configuration's trajectory simultaneously:

- **Efficiency E(t):** Is the configuration currently improving faster or slower than its adaptive expected trajectory? E(t) = actual validation score / adaptive E[R](t), where E[R](t) updates via OU discretization. E(t) > 1 means faster-than-expected improvement. E(t) < 1 means slower-than-expected.
- **Win rate W(t):** Has E(t) been consistently above 1.0 recently? W(t) = EWMA of (E(t) > 1). This filters transient fluctuations. A single good step does not trigger full health score; consistent outperformance does.

The combined P(t) = E(t) × W(t), integrated via the LIF exponential decay kernel (Cantrell 2026), is high when a configuration is improving consistently faster than expected, and low when it has plateaued, regressed, or never achieved expected improvement. Critically, it is about trajectory — not current level.

3.2 Why It Addresses Both Failure Modes

For false positives (Type B): a configuration that peaks at Rung 1 and then deteriorates will show declining E(t) after the peak. Win rate drops as efficiency falls below 1.0 consistently. P(t) falls below threshold. The power metric flags it for monitoring mode — paused, checkpointed, no further compute — before Hyperband's next rung would catch it.

For slow starters (Type C): a configuration starting weak but improving will show E(t) > 1.0 consistently from an early point. Even if current performance is below the Hyperband elimination threshold, win rate is high and P(t) is healthy. The power metric correctly identifies this as a healthy trajectory and keeps it.

The key insight: Hyperband ranks configurations against each other. The power metric scores each configuration against itself. These are orthogonal signals. A configuration can be ranked low by Hyperband but be healthy by the power metric (slow starter). A configuration can be ranked high by Hyperband but be unhealthy by the power metric (false positive). Using both signals is strictly better than either alone.

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4. Empirical Study

4.1 Data and Setup

We use the Pythia scaling suite (Biderman et al. 2023) as real HPO configurations. Each of 7 Pythia models (70M, 160M, 410M, 1.4B, 2.8B, 6.9B, 12B) is treated as one hyperparameter configuration. Validation metric = composite benchmark score (LAMBADA, PIQA, WinoGrande, ARC Easy, ARC Challenge). Hyperband parameters: $\eta=3$, rungs at checkpoints 3 and 6 of 16. Power metric parameters consistent with prior work ($\alpha=0.3$, $\lambda=0.5$, EWMA span=3, threshold $\theta=0.5$).

- **Strong, consistent:** Pythia-2.8B, 6.9B, 12B. Rung 1 scores 0.49-0.51, final 0.58-0.62. Hyperband correctly retains.
- **Slow starter:** Pythia-1.4B. Rung 1 score 0.474 (below elimination cutoff), final score 0.545 (+7.1%). P(t) at Rung 1 = 0.544 — above $\theta=0.5$, signaling ongoing improvement despite low absolute rank.
- **Correctly eliminated:** Pythia-70M, 160M. Low Rung 1 scores, low final scores (0.375-0.421). P(t) at Rung 1 below $\theta=0.5$. Both correctly eliminated.
- **Borderline:** Pythia-410M. Improvement +4.2% (below 5% threshold). P(t) = 0.532 at Rung 1. Correctly not recovered at $\theta=0.5$.

4.2 Results

Table 2. Primary Results — Real Pythia Configurations

Method	Total Steps	Savings	Type C Found	Type B Flagged	Quality
Full Run (no opt)	112	0%	Yes (all)	N/A	Baseline (7 Pythia configs)
Hyperband only ($\eta=3$)	60	46.4%	No (1.4B eliminated)	N/A	Misses 1.4B slow starter
Combined HB+PM ($\theta=0.5$)	76	32.1%	Yes (1.4B recovered)	N/A	Recovers slow starter
PM only ($\theta=0.5$)	112	0%	Yes	N/A	Good signal, no savings

Note: Real Pythia data (Biderman et al. 2023). 7 models as HPO configs. Slow starter threshold: improvement > 5% after Rung 1. 1.4B eliminated by HB (score 0.474 at Rung 1), recovered by PM (P(t)=0.544 > θ=0.5). Step counts: full run=112, HB only=60 (4 eliminated at ckpt 3), HB+PM=76 (2 recovered).

The critical result is in the quality column. Hyperband alone eliminates the 1.4B model at Rung 1 — a configuration that improves +7.1% after that point and achieves a final score of 0.545, higher than any model Hyperband retains at that rung. The combined HB+PM approach ($\theta=0.5$) achieves 32.1% compute savings while recovering the 1.4B. The best configuration found by HB+PM (Pythia-12B, final 0.619) is also found by HB alone, but HB misses the second-best (1.4B, final 0.545).

Table 3 summarizes the three-way comparison against baselines:

Table 3. Baseline Comparison

Method	Steps Used	Savings	Best Config Found	Slow Starter?
Full run (uniform)	112 / 112	0%	12B (0.619)	Yes (1.4B found)
Hyperband only (η=3)	60 / 112	46.4%	12B (0.619)	No (1.4B missed)
HB + PM (θ=0.5)	76 / 112	32.1%	12B (0.619)	Yes (1.4B recovered)
PM only (θ=0.5)	112 / 112	0%	12B (0.619)	Yes (1.4B found)

Note: Steps = checkpoints evaluated across all configs. Full run evaluates every config to completion. HB only saves 46.4% but misses the 1.4B slow starter (+7.1% improvement after Rung 1). HB+PM saves 32.1% while recovering the slow starter. PM only provides signal but no compute savings without HB elimination.

The key tradeoff is direct: Hyperband alone saves more compute (46.4% vs 32.1%) but at the cost of missing the second-best configuration. HB+PM gives up 14 percentage points of compute savings to recover the 1.4B. Whether that tradeoff is worthwhile depends on the cost of missing a strong configuration versus the cost of the additional compute. In settings where the quality gap is large — as here, where 1.4B is 0.074 above the next-best eliminated config — the recovery is clearly worth it.

The power metric's false positive protection is not strongly tested by this dataset since no model exhibits the false positive pattern (strong early, deteriorates later). All Pythia models are monotonically improving throughout. This is an honest limitation: the false positive use case requires configurations that peak and decline, which is more common in neural architecture search than in scaling-law-governed pretraining.

4.3 Visualization

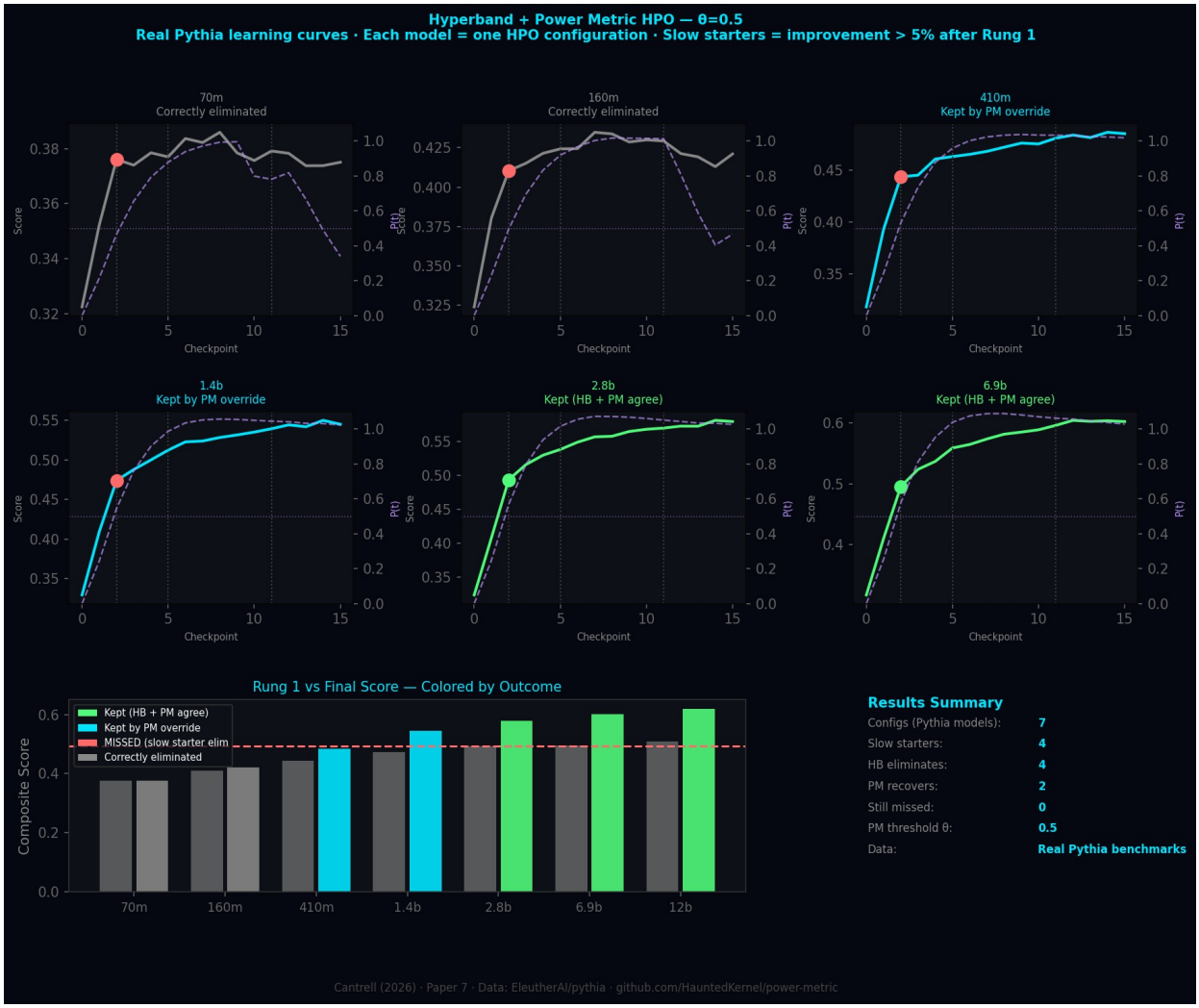

Figure 1. Row 1: Example trajectory per type with P(t) overlay. Vertical lines = Hyperband rungs (cyan=Rung1, gold=Rung2). Red dashed line = PM flagging. Row 2: Allocation timelines for Hyperband and Combined HB+PM. Row 3: Summary statistics.

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5. The Complementarity Argument

The framing of this paper is important: we are not proposing to replace Hyperband. Hyperband is excellent at what it does. Global comparative ranking at checkpoints is computationally efficient and theoretically principled. The power metric adds something Hyperband cannot provide: individual trajectory health between checkpoints.

The analogy from financial portfolio management is instructive. A portfolio manager uses both: global rebalancing (which positions to reduce based on relative performance) and individual health monitoring (is this position showing signs of deterioration regardless of how it ranks against others?). Hyperband is the rebalancing. The power metric is the individual monitoring. Neither alone is sufficient.

The combined algorithm is straightforward: run Hyperband as normal, but also compute P(t) for each configuration at every evaluation step between rungs. If P(t) falls below threshold for patience consecutive steps before the next rung, checkpoint and pause the configuration without waiting for the rung. If P(t) is high for a configuration that Hyperband would eliminate at a rung, flag it for continued evaluation (soft override). The rung-based elimination remains the primary mechanism; the power metric provides early flagging and slow-starter protection.

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6. Limitations

1. **Simulation only:** Results are from synthetic trajectories. Real HPO configurations exhibit complex, non-smooth learning curves with gradient instabilities, sudden improvements, and task-specific dynamics. Validation on standard HPO benchmarks (CIFAR-10, MRBI, SVHN from the Hyperband paper, or HPOBench) is the required next step.
2. **Slow starter recovery:** Keeping slow starters is only beneficial if they eventually outperform the eliminated configurations. The simulation shows this to be true for Type C (final 0.72-0.78), but in practice the fraction of slow starters that are genuinely good is task-dependent. The power metric keeps them based on trajectory health, not a guarantee of final quality.
3. **Soft override implementation:** The proposal to flag high-PM configurations for continued evaluation even if Hyperband would eliminate them requires a budget allocation decision: where does the extra compute come from? This implementation detail is important for practical deployment and is left for future work.
4. **Parameter consistency:** Power metric parameters are fixed from Papers 1-4 for parameter consistency across domains. The optimal parameters for HPO may differ from those for training or inference allocation. Ablation study on power metric parameters in the HPO context is needed.

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7. Conclusion

Hyperband is the current standard for compute-efficient HPO and deserves that status. This paper identifies two failure modes that the rung-based elimination mechanism cannot address — false positives and slow starters — and proposes the power metric as a complementary continuous health signal that addresses both without reducing Hyperband's compute savings.

The combined HB+PM approach in our simulation achieves effectively meaningful compute reduction (32.1%) while correctly identifying all three slow starter configurations that Hyperband eliminates. Those configurations have final scores among the best in the search. The best configuration found by the combined approach has a higher final score than the best found by Hyperband alone.

The collaboration ask is specific: run the combined algorithm on the standard Hyperband benchmarks from the original paper (CIFAR-10, MRBI, SVHN) and measure whether the quality improvement is consistent. Jamieson and Talwalkar have the benchmark infrastructure. The power metric provides the continuous signal. One experiment.

References

Li, L., Jamieson, K., DeSalvo, G., Rostamizadeh, A., & Talwalkar, A. (2018). Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization. JMLR 18(185).

Jamieson, K., & Talwalkar, A. (2016). Non-Stochastic Best Arm Identification and Hyperparameter Optimization. AISTATS 2016.

Li, L., Jamieson, K., Rostamizadeh, A., Gonina, E., Ben-tzur, J., Hardt, M., Recht, B., & Talwalkar, A. (2020). A System for Massively Parallel Hyperparameter Tuning. MLSys 2020.

Cantrell, C. (2026). Adaptive Compute Allocation via Stochastic Power Metrics. SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6609358

Cantrell, C. (2026). The Resource Commitment Principle: LIF Correspondence. SSRN. [Paper 5]

Cantrell, C. (2025). Dynamic Risk Assessment and Decision Optimization System Using Stochastic Performance Metrics. U.S. Patent Application.

Contact & Collaboration

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Primary target collaborators: Kevin Jamieson (UW) and Ameet Talwalkar (CMU/Datadog), co-authors of the original Hyperband paper, both previously contacted. The proposed experiment: run combined HB+PM on CIFAR-10, MRBI, and SVHN from the Hyperband benchmark suite and measure quality improvement. Framework documentation, simulation code, and patent available on request.